



Course Title:	Calculus
Course Code:	SMTH-101
Semester:	1
Credit Hours:	3 Credit Hours
Pre-requisites:	None

Learning Outcomes

By the end of this course, students will be able to:

1. The students will be able to understand the concept of limit and continuity of the function fundamental concepts of the derivative (in terms of tangential slope and instantaneous rate), and integration, (in terms of accumulated change and area under the curve)
2. The students will be able to apply various techniques to compute derivatives and integrals of different function types used in optimization problems and other related applications.
3. The students will be able to understand the concept of multi variables functions, partial derivative and double integration.

Course Outline

Unit 1

Introduction to Calculus

Functions, Rule of Four, independent variable, dependent variable, domain, and range of a function, Toolkit Functions: linear, quadratic, polynomial, rational, transcendental functions- Introduction to Limit and continuity: Intuitive and Geometric Approach, techniques of finding limits, Continuous and discontinuous functions, Slope of a function, Average rate of change, Secant Lines

Unit 2

Concept and idea of differentiation

Instantaneous rate of change, Tangent line, Geometrical and Physical meaning of derivatives, Rules of differentiation, Techniques of Differentiation, Product rule, quotient rule, chain rule Applications of derivative:

Increasing decreasing function, Local Maxima and Minima, critical points of a function, First derivative test, Inflection point, Concavity, Second derivative test, Newton Raphson method

Unit 3

Introduction to Integration

The indefinite integral of exponential, polynomials, and trigonometric functions, Techniques of integration (by substitution, by parts), Techniques of integration (by partial fraction), Integration as Area under the curve, Definite Integral, properties of definite integral, Fundamental theorem of calculus, Area between two curves, Trapezoidal Rule, Functions of several variables, Derivatives of several variables, Multiples integrals

- **Teaching-learning Strategies:**
Class Lecture method, which includes seminars, discussions, assignments and projects. (Audio-visual tools are used where necessary)
- **Assignments-Types and Number with calendar:**
According to the choice of respective teacher.
- **Assessment and Examinations:**
According to the University's Semester Rules.

Sr. No.	Elements	Weightage	Details
1	Midterm Assessment	35%	It takes place at the mid-point of the semester.
2	Formative Assessment	25%	It is continuous assessment. It includes: Classroom participation, attendance, assignments, and presentations, homework, attitude and behavior, hands-on-activities, short tests, quizzes etc.
3	Final Assessment	40%	It takes place at the end of the semester. It is mostly in the form of a test, but owing to the nature of the course the teacher may assess their students based on term paper, research proposal development, field work and report writing etc.

Text Books

1. APPLIED CALCULUS by Hughes Hallett (4th Edition)
2. Thomas Calculus by G B Thomas, 11th Edition.

Suggested Readings

1. CALCULUS by Howard Anton 10th Edition
2. CALCULUS AND ITS APPLICATION by Marvin L. Bittinger 10th Edition
3. Calculus single and Multivariable by Hughes Hallett al., 6th edition.
4. James Stewart, "Multivariable Calculus" 7th Ed.



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Give short answer to the following concepts:

(6x5=30)

(a) Evaluate the integral $\int 3x\sqrt{x^2+1}dx$.

(b) Position function of the particle is, $s = 1 + 5t - 2t^2$ (where s is in meters and t is in seconds). Find average velocity over the interval $[2, 3]$ and velocity at $t = 2$.

(c) For what value of 'a', function $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 3 \\ 2ax & \text{if } x \geq 3 \end{cases}$ is continuous at $x = 3$.

(d) Express $3 < x < 7$ in modulus notation.

(e) Evaluate $\int_2^{\infty} \frac{1}{x(\ln x)^3} dx$.

(f) Find $\lim_{x \rightarrow 3} \frac{3-x}{|x-3|}$.

Answer the following questions.

(5x6=30)

Q. 2. If $y = (\sin^{-1} x)^2$ then prove that $(1-x^2)y^{(n+2)} - (2n+1)xy^{(n+1)} - n^2y^{(n)} = 0$.

Q. 3. Use Mean Value Theorem to show that $|\tan x + \tan y| \geq |x + y|$ for all real numbers x and y in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Q. 4. Solve the inequality $x^2 - x - 2 \geq 0$.

Q. 5. Find the asymptotes of the graph $y = \frac{x^2-3}{2x-4}$.

Q. 6. Evaluate $\int \frac{1}{\tan x - \sin x} dx$.

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Q.1. Solve the following:

(6x5=30)

(i)	Solve the inequality : $\left \frac{2}{x} - 4\right < 3$.
(ii)	Find the limit of $\lim_{x \rightarrow 1} \frac{1}{ 1-x }$
(iii)	Discuss the continuity of function $f(x) = \frac{x^4+20}{5x(x-2)}$ at $x = 2$.
(iv)	Find the Maclaurin series for $\frac{1}{1-x}$.
(v)	Find domain of the function. Check whether the function is either even or odd. $f(x) = x^2 + \sec(2x) - 1$
(vi)	Evaluate the Integral $\int (x^{9/4} - x^{-1/8} - 5) dx$.

Solve the following:

(5x6=30)

Q.2	Solve the integral $\int e^{2+x^3} x^5 dx$.
Q.3	Differentiate $y = \frac{\cos x}{1-\sin x}$
Q.4	Find the reduction formula of $\int \sec^n x dx$ and use it to evaluate $\int \sec^3 x dx$.
Q.5	Find the limit of $\lim_{t \rightarrow 1} \frac{t-1}{\sqrt{t}-1}$
Q.6	Discuss the continuity and differentiability of $f(x) = x $ at $x = 0$.



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Q.1. Solve the following:

(6x5=30)

(i)	Does the limit of $f(x) = \frac{ x }{x}$ exist at $x = 0$	(5)
(ii)	Show that $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ Is differentiable at $x = 0$ and find $f'(0)$	(5)
(iii)	Solve the inequality $\frac{6-x}{4} < \frac{3x-4}{2}$	(5)
(iv)	Find the function $f(x)$ whose derivative is $\sin x$ and whose graph passes through the point $(0, 2)$.	(5)
(v)	Find $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x}{1 + \tan x}$	(5)
(vi)	Find $\frac{dy}{dx}$ if $y = \int_x^5 3t \sin t dt$	(5)

Solve the following:

(5x6=30)

Q.2	Draw the graph of $\sin x$. Using the properties of scaling and shifting draw the graph of $y = 1 + \sin(x + \frac{\pi}{4})$.	(6)
Q.3	Evaluate $\left(\frac{-2+i\sqrt{3}}{\sqrt{5}-6i}\right)^3$	(6)
Q.4	For what value or values of constant m , if any, is	(6)
	$f(x) = \begin{cases} \sin 2x, & x \leq 0 \\ mx, & x > 0 \end{cases}$ a) Continuous at $x = 0$? b) Differentiable at $x = 0$?	
Q.5	Find absolute maximum and minimum values of $f(x) = x^{\frac{2}{3}}$ on the interval $[-2, 3]$.	(6)
Q.6	Find the area of the region bounded by x -axis and the graph of	(6)
	$f(x) = x^3 - x^2 - 2x, \quad -1 \leq x \leq 2$	



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : First Semester – Fall 2021

Roll No.

Paper: Mathematics A-I [Calculus(I)]
Course Code: MATH-101 / MATH 11010

Time: 3 Hrs. Marks: 60

Q.1. Solve the following:

(6x5=30)

(i)	Does the limit of $f(x) = \frac{\cos x}{1-x}$ exist at $x = \pi$	(5)
(ii)	Solve $z^4 - 3z^2 + 2 = 0$, where z is a complex number.	(5)
(iii)	Solve the inequality $\left \frac{2}{x} - 4 \right < 3$	(5)
(iv)	Find the value or values of c that satisfy the mean value theorem for the function $f(x) = \sqrt{x-1}$ in $[1,3]$	(5)
(v)	Find $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$.	(5)
(vi)	At what point the function is continuous: $f(x) = \frac{x \tan x}{x^2 + 1}$	(5)

Solve the following:

(5x6=30)

Q.2	Graph the function and comment on continuity and differentiability at $x=1$ $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases}$	(6)
Q.3	Discuss the continuity of the following function at $x=2$ and $x=-2$ $g(x) = \begin{cases} \frac{x^2-8}{x^2-4}, & x \neq 2, -2 \\ 3, & x = 2 \\ 4, & x = -2 \end{cases}$	(6)
Q.4	Evaluate $\int_0^{\pi/2} \frac{\cos \theta \, d\theta}{(3+2\sin \theta)(1-\sin \theta)}$.	(6)
Q.5	For what values of a, m and b does the function $g(x) = \begin{cases} 3, & x = 0 \\ -x^2 + 3x + a, & 0 < x < 1 \\ mx + b, & 1 \leq x \leq 2 \end{cases}$ Satisfy the mean value theorem on $[0,2]$.	(6)
Q.6	Find the volume of solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y=1$, $x=4$, about the line $y=1$.	(6)